GM	WORLDWIDE ENGINEERING STANDARDS	Test Procedure Chassis	GMW15355

Tire Rapid Air Loss Test Procedure

1 Scope

Note: Nothing in this standard supersedes applicable laws and regulations.

Note: In the event of conflict between the English and domestic language, the English language shall take precedence.

1.1 Purpose. The purpose of this procedure is to define a test method for determining if a rim can retain a deflated tire in the event of a rapid loss of inflation pressure from 96.5 km/h (60 mph), until the vehicle can be stopped with a controlled braking application.

1.2 Foreword.

1.2.1 European hard metric tire size designations are judged to be equivalent to P-metric size designations and visa versa for the purpose of this test.

1.2.2 Tire speed rating and tire type can be ignored for the purpose of this test.

1.2.3 The rim must be approved for usage with the release tire size in accordance with S4.4 of FMVSS 571.109.

1.2.4 For the purposes of this test, rim widths within 12.7 mm (0.5 in) are judge to be equivalent.

1.2.5 The vehicle maximum load on the tire shall not be greater than the applicable maximum load rating as marked on the sidewall of the tire.

1.2.6 The rim must retain the tire in the event of rapid loss of air until the vehicle can be stopped from 96.5 km/h (60 mph) by controlled braking.

1.2.7 The tire deflation system should be capable of deflating the tire in one (1) s or less.

1.3 Applicability. This test procedure applies to tires and compact spares for passenger cars and light trucks with a Gross Vehicle Weight Rating (GVWR) of 4536 kg (10 000 lb) or less.

2 References

Note: Only the latest approved standards are applicable unless otherwise specified.

2.1 External Standards/Specifications.

FMVSS 571.109 FMVSS 571.110

2.2 GM Standards/Specifications.

None

3 Resources

3.1 Facilities. This test is run on a straight level asphalt surface with a grade not to exceed 1% in any direction and sufficient to achieve the required test speed and that is clear of debris.

3.2 Equipment.

3.2.1 Use a scale of sufficient capacity to measure maximum weight on either front or rear axle. A print out of the vehicle weight should be acquired.

3.2.2 Full driver safety equipment per test location requirements.

3.2.3 Calibrated tire inflation gage.

3.2.4 Equipment for measurement of deceleration.

3.2.5 Equipment for tire deflation.

3.2.6 Sandbags or other ballast.

3.2.7 Use calibrated road speed measurement equipment, such as fifth wheel and associated speed readout. The test vehicle may be equipped with a calibrated speedometer as an alternative.

3.3 Test Vehicle/Test Piece.

3.3.1 Test Vehicle. A vehicle equipped to the extent necessary (and as prescribed by the requesting test engineer) to perform the test and loaded to the equivalent of maximum tire load capacity.

3.3.2 Test Piece. The test tire and wheel sizes are specified by the test requestor.

3.3.2.1 A new tire and wheel should be used which are confirmed as being within production tolerance by the test requestor/component supplier. Two tire and wheel assemblies are required for the rapid loss of air test.

3.4 Test Time.

Calendar time:	0.5 day
Coordination hours:	1 hour
Setup hours:	3 hours
Test hours:	1 hour

Note: Times are for testing two (2) samples.

3.5 Test Required Information. All test required information shall be provided by the test requestor.

3.6 Personnel/Skills. Personnel must comply with local driver regulations and certifications.

4 Procedure

4.1 Preparation.

4.1.1 Measure wheels to confirm they meet specifications.

4.1.2 Complete the Metrology Data Sheet Appendix A.

4.1.3 Record all required information on the Test Setup Data Sheet Appendix B.

4.1.4 Mount each tire to its test rim as stated on the test request with the vehicle manufacturer's recommended lubricant.

4.1.5 Set all tire air pressures as stated on the test request.

4.1.6 Ballast the vehicle to the requested weight and weight distribution, such that the test wheel is equal to the test load specified in on the test request.

Note: All equipment and ballast inside the passenger compartment of the vehicle must be securely held in place. External ballast box and a metal retaining screen installed between the cargo compartment and the passenger cab. Remove all loose equipment from the passenger compartment including tools, spare parts, and personal belongings.

4.1.7 Calibrate speedometer or install a suitable external speed measuring device.

4.1.8 Install or set up the tire deflation system.

4.2 Conditions.

4.2.1 Environmental Conditions.

4.2.1.1 The test should be conducted on a dry paved surface.

4.2.1.2 The test should be performed with an ambient temperature between 0 and 38°C (32 and 100°F).

4.2.1.3 The test should be performed with a wind velocity not to exceed 5 m/s (16 ft/s).

4.2.2 Test Conditions. Deviations from the requirements of the procedure shall have been agreed upon. Such requirements shall be specified on component drawings, test certificates, reports, etc.

4.3 Instructions.

4.3.1 Accelerate to and maintain a vehicle speed of 96.5 +0, -2 km/h (60 +0, -1 mph) and simulate the rapid loss of air (Time $\leq$ 1 s) from the tire, through an opening at least equal to a 1.3 cm (1/2 in.) diameter hole.

4.3.2 Upon initial release of air form the tire, apply the brakes to bring the vehicle to a controlled stop using the most rapid constant deceleration rate not exceeding 2.5 m/sec^2 (8 ft/sec^2), without wheel lock.

4.3.3 Evaluate retention of tire to wheel, paying particular attention to the location of the tire beads with respect to the wheel flanges.

4.3.4 Record the information on the Results Data Sheet Appendix C.

5 Data

5.1 Calculations. Not applicable.

5.2 Interpretation of Results. The standard requires that the rim retain the deflated tire until the vehicle can be stopped. Retain the **deflated tire** means that the tire beads must be within the rim and, at no time shall a tire bead be outside the rim so that the rim is operating on the inside of the deflated tire.

5.3 Test Documentation. The corresponding forms shall be completed. See Appendix A thru C.

- Metrology Data Sheet
- Test Setup Data Sheet
- Results Data Sheet

6 Safety

This standard may involve hazardous materials, operations, and equipment. This standard does not propose to address all the safety problems associated with its use. It is the responsibility of the user of the method to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

7 Notes

7.1 Glossary. None.

7.2 Acronyms, Abbreviations, and Symbols.

GVWR Gross Vehicle Weight Rating

8 Coding System

This standard shall be referenced in other documents, drawings, etc., as follows:
Test to GMW15355

9 Release and Revisions

9.1 Release. This standard originated in May 2006, replacing GMUTS R-10B-110G, MTL9056, and ON 5230.9. It was first approved by the Tire

Global Subsystem Leadership Team in June 2006.
It was first published in October 2007.

Appendix A

RAPID AIR LOSS
(Metrology Data Sheet)

TEST REQUEST NUMBER ____________ DATE _____________
WHEEL PART NUMBER ______________
WHEEL SIZE ________ x ________
RIM CONTOUR ________________
OFFSET _____________________
INSPECTOR __________________

WHEEL CHARACTERISTICS INSPECTED

A. RIM FLANGES
1. Height HIGH AND LOW
2. Width MINIMUM
3. Radius MINIMUM
4. Diameter OUTBOARD
5. Diameter INBOARD

B1. RIM WIDTH

C. RIM BEAD SEATS
1. Angle NOMINAL
2. Width MINIMUM
3. Heel Radius MAXIMUM
4. Diameter OUTBOARD - BALL TAPE
5. Diameter INBOARD - BALL TAPE

D. RIM WELL CLEARANCE
1. Maximum (L4) Dimension

E. RIM HUMPS
1. Location MAXIMUM AND MINIMUM
2. Shape MAXIMUM
3. Radii MAXIMUM

F1. THE RIM CONFORMS TO THE T&RA CONTOUR GAGE AND THE ABOVE STANDARDS

RAPID AIR LOSS TESTING RESULTS
A = APPROVED R = REJECT

WHEEL ID #	A1	A2	A3	A4	A5	B1	C1	C2	C3	C4	C5	D1	E1	E2	E3	F1

Appendix B

Rapid Air Loss Test

(Test Setup Data Sheet)

Test Location

Test Region		Test Facility		Test Road	

Equipment Information

Vehicle Speed (Odo. Other)		Air Pressure Gauge		Tire Deflation Method
Device Type		Device Type		Device: ____________________
Calibration Date		Calibration Date		___________________________

Vehicle Information

Vehicle ID		Vehicle Type (Check Box)	
Make		Production	[ ]
Model		Prototype	[ ]
VIN Number		Other	[ ]

Test Component Information

Tire Information	Tire ID	_____________	Tire ID	_____________
Manufacturer				
Size				
Construction Number				
Part Number				
Load Range				

Wheel Information	Wheel ID	_____________	Wheel ID	_____________
Manufacturer				
Size				
Rim Contour				
Part Number				
Material				

Tire Test Load Information (Measured)

Tire Test Load (Kg)	Tire Test Position on Vehicle	Tire Air Pressure (kPa)

Appendix C

Rapid Air Loss Test

(Results Data Sheet)

Test Results

	Wheel ID	___________	**Wheel ID**	___________
Was tire retained with both beads inside the rim?	**Yes** [ ]	**No** [ ]	**Yes** [ ]	**No** [ ]
Was the vehicle stopped with a controlled brake application (driver opinion)?	**Yes** [ ]	**No** [ ]	**Yes** [ ]	**No** [ ]
Was the deflation opening in the tire greater than 1.3 cm?	**Yes** [ ]	**No** [ ]	**Yes** [ ]	**No** [ ]

Ambient Temperature: _______________________°C.
Maximum Deceleration Rate: __________________m/sec^2.

Tested by ____________________________________ **Date** ______________________
(Print)

Comments: ____

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Requestor Approval __________________________ **Date** ______________________
(Signature)

Comments ____

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